# A quartic in 370 variables with specified integer coefficients

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**Abstract**

We specify an integer quartic in 370 total variables whose rational input projection is the integers, under the arithmetic premises of the specified elliptic source construction. The coefficient table contains 5837 nonzero monomials. The construction uses three height detectors applied to rational linear forms in a sum of four elliptic abscissas. An effective bound for the height lost in that sum supplies an explicit numerical parameter. All four point charts allow arbitrary nonzero integer indices. The resulting quartic has content one and a largest coefficient with 1791357 binary digits. We also specify a family with the same defining property for every integer parameter above an explicit threshold.

## 1 The polynomial and its arithmetic premises

For a polynomial $F \in \mathbb{Z}[X, Y_1, \dots, Y_m]$, its rational input projection is the set

$$\{x \in \mathbb{Q} : \exists (y_1, \dots, y_m) \in \mathbb{Q}^m,\ F(x, \boldsymbol{y}) = 0\}.$$

All variable counts include $X$. Degree is total degree. The height $H(F)$ of an integer polynomial is its largest absolute coefficient. For a rational or algebraic number, $h$ denotes absolute logarithmic Weil height.

Put $\ell = 80989$. The exact tables `quadratic-family370.json` and `polynomial-family370.json` specify

$$A_N = \ell^{2N+1}, \qquad P_N = \sum_{j=1}^{342} Q_{N,j}^2 \in \mathbb{Z}[X, Y_1, \dots, Y_{369}],$$

where every $Q_{N,j}$ is quadratic. The table records the powers of $A_N$ separately to keep large integer coefficients readable; $A_N$ is a coefficient, not a variable.

**Theorem 1.1.** *Assume the arithmetic premises stated below. For every integer $N \geq 27464$ and every $x \in \mathbb{Q}$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \boldsymbol{y} \in \mathbb{Q}^{369},\ P_N(x, \boldsymbol{y}) = 0.$$

*Every such polynomial has degree exactly four, content one, and 5837 nonzero monomials. Its coefficient height satisfies*

$$H(P_N) < 2^{1520}\, 80989^{4N+2}.$$

*In particular, the single polynomial $P_{27464}$ is fully specified by substituting $A = 80989^{54929}$ in the table. Its largest coefficient has exactly 1791357 binary digits.*

We retain the arithmetic contract of the explicit source in [1, Section 2]. In particular, with

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad d = 3 - \rho, \quad E : y^2 = x^3 - d^2 x,$$

$$R_0 = (1 + \rho, 2\rho), \qquad P = 2R_0 = \left( \frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32} \right),$$

the group and index-height premises are

$$2E(K) = \mathbb{Z}P, \qquad h(x([n]P)) \geq n^2 - 6 \quad (n \in \mathbb{Z} \setminus \{0\}). \tag{1}$$

The contact-height premise, pole obstruction and simultaneous pole realization, and simultaneous prime-pattern premise are unchanged. The quantitative weighted selection premise used in the source's real-separation argument is also retained. These inputs are explicitly enumerated in [1, Section 2]; coefficient verification does not establish them.

For reference, the contact-height constant is

$$\mathcal{H} = 2^{110}210^{1800}\left(\prod_{p\leq 76176} p\right)^{19} + 128 < 2^{2093448}. \tag{2}$$

The effective abscissa-sum theorem used below is proved separately in [4]. It is not an additional conjectural premise, and it does not require an ineffective height constant.

## 2 Point charts and the three height scales

The four root charts retain the quadratic equations over $K$

$$x = h^2, \qquad j^2 = x^2 - d^2. \tag{3}$$

The corresponding elliptic point is $Q = (x, hj)$.

**Lemma 2.1.** *Under (1), each chart (3) gives a point $Q = [m]P$ with $m$ a nonzero integer. Conversely every nonzero integer multiple of $P$ admits such a chart. No parity restriction on $m$ is needed.*

*Proof.* The field $K$ is real, so $h = 0$ would imply $j^2 = -d^2$, an impossibility. Thus the chart gives a finite nonzero point of $E$. By (1), write $Q = mR_0 + T$ for $T \in E[2](K)$. The standard two-isogeny squareclass homomorphism has value $[x(Q)]$ away from the origin and $(0,0)$, value 1 at the origin, and value $[-d^2]$ at $(0,0)$. Its values on $E[2](K)$ are $1, -d^2, d, -d$. These four classes have uniform signs at the two real embeddings, whereas $[x(R_0)] = [1+\rho]$ has mixed signs. Since $x(Q)$ is a square, $m$ is even. The torsion class must then be a square. Neither negative class is a square, and $d$ is not a square because its rational norm is 7. Hence $T = O$ and $Q = [m/2]P$; finiteness makes this index nonzero.

Conversely every such point is a double. The duplication formula

$$x(2V) = \left(\frac{x(V)^2 + d^2}{2y(V)}\right)^2$$

gives $x = h^2$, and putting $j = y/h$ gives the second chart equation. The relevant numerator and denominator cannot vanish for a nontorsion point. The squareclass argument and all exceptional point exclusions are also detailed in [3, Section 2]. □

Write the four root points as

$$Q_1 = [n]P, \qquad Q_2 = [a]P, \qquad Q_3 = [t]P, \qquad Q_4 = [b]P, \qquad B = \max(|n|, |a|, |t|, |b|).$$

Define

$$S = x(Q_1) + x(Q_2) + x(Q_3) + x(Q_4) = S_0 + S_1\rho, \qquad S_0, S_1 \in \mathbb{Q}.$$

Every abscissa is positive at both real embeddings. In fact its curve equation and its nonzero ordinate give $x(Q_i) > d$ at each embedding. Consequently $S_0 > 12$ and $S_0 > \sqrt{2}|S_1|$. In particular, both $S_0$ and $S_0 + S_1$ are positive.

**Lemma 2.2.** *For every tuple supplied by the root charts,*

$$B^2 \le 2h(S_0) + 2h(S_0 + S_1) + 82.$$

*Proof.* The effective sum-height theorem [4] states

$$h(S) \ge \max_i h(x(Q_i)) - 24 \log B - \log 864.$$

Its proof compares the integral local principal parts $x([n]P) = (nL_v)^{-2} + O(1)$ at each finite place. For a fixed target index, only subsets containing that index can cause loss; their positive integer numerators have total size bounded by $B^{24} \cdot 864$. Positivity prevents loss at the real places. Applying (1) and the elementary logarithm bound gives $B^2 \le 2h(S) + 80$. Write $U = S_0$ and $V = S_0 + S_1$, so

$$S = (1 - \rho)U + \rho V.$$

Both coefficients are algebraic integers. Hence the finite-place part of $h(S)$ is at most that of $h(1, U, V)$. At the two real embeddings, the sums of their absolute values are $2\sqrt{2} - 1$ and $2\sqrt{2} + 1$, whose product is 7. Therefore

$$h(S) \le h(1, U, V) + \tfrac{1}{2} \log 7 \le h(U) + h(V) + \tfrac{1}{2} \log 7.$$

Since $\log 7 < 2$, the assertion follows. □

Use the three positive rational square scales

$$1, \qquad S_0^2, \qquad (S_0 + S_1)^2. \tag{4}$$

All point tests, logarithm approximations, comparison tests and pole equations in the three detectors retain their exact source forms. Their orders are $1, 10, 40$. The local nonvanishing argument of [2] applies to arbitrary nonzero square scales, including these three scales; it covers the degenerate input $t = 1$ and empty contact radicals. Thus no deleted inverse witness is reintroduced.

**Lemma 2.3.** *Suppose $J$ is a nonzero integer and the $k$-th powers of all three contact radicals divide $J$. Then*

$$B^2 \le 5\mathcal{H}|J|^{19/k}.$$

*Proof.* Put $T = (\mathcal{H} + \log 3)|J|^{19/k}$. The retained contact-height theorem bounds each of

$$h(\lambda), \quad h(S_0^2\lambda), \quad h((S_0 + S_1)^2\lambda)$$

by $T$; its fractional linear contact change accounts for $\log 3$. Since the detector domain excludes $\lambda = 0$,

$$h(S_0) \le T, \qquad h(S_0 + S_1) \le T.$$

Lemma 2.2 now gives

$$B^2 \le 4T + 82.$$

Here $|J| \ge 1$ and the explicit constant (2) is greater than 92. Therefore $4T + 82 \le 5\mathcal{H}|J|^{19/k}$. □

## 3 The explicit scalar parameter and soundness

The input label and scalar polynomial are

$$\beta = 51X^2 + 2504, \qquad r = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1),$$

$$R = \frac{A_N r}{1+\ell^2 r}, \qquad c_R = R + 9792.$$

The retained scalar block, with the indicated shift, is precisely the parity-free scalar block of [3, Sections 4–5]. Its integral group identity is $[m](2G) = [4(a-1)]H$; it does not require $a-1$ to be even. The existing exceptional-point exclusions and denominator guards are retained.

For clarity, its conclusions before any height comparison are

$$c := |a-1| \geq \ell^{L_N}, \qquad L_N = \left\lfloor \frac{187(2N-1)}{8} \right\rfloor + 1, \qquad h(\beta) \leq \mathcal{H} c^{19/24}. \tag{5}$$

The proof uses the same pole obstruction, the same formal comparison and the same five contact values as the source. The fixed prime avoids these five squareclasses: for each $\theta \in \{0, 1, 2401, -343/32, 1/2\}$, the rational number $(\theta - 2504)/51$ is a nonsquare modulo 80989. The exact Euler tests are repeated by the checker.

At $N = 27464$ one has $L_N = 1283919$. The integer certificates

$$80989^{4096} > 2^{66787}, \qquad 66787 \cdot 1283919 - 4096 \cdot 20934550 = 1181453 > 0$$

and (2) imply, for every $N \geq 27464$,

$$c > 2^{20934550} > (80\mathcal{H})^{10}, \qquad |a| > (40\mathcal{H})^{10}, \qquad \log(48\mathcal{H}) < |a|/40, \qquad h(\beta) < |a|/40. \tag{6}$$

These are the same elementary numerical implications as in [1, Section 4]. For example, $c^{5/24} > 80\mathcal{H}$ gives $h(\beta) < c/80 \leq |a|/40$, and the remaining estimates follow directly from $c - 1 \leq |a|$.

The unchanged comparison equations imply the following divisibilities. Put

$$\Delta = t^2 - a(a-2)tb + (a-2)^2 b^2 - 1.$$

The fortieth powers of all three contact radicals divide $\Delta$. If $n - \beta \neq 0$, their tenth powers divide the nonzero reduced numerator $J$ of $n - \beta$. These are the local comparison conclusions of the source; removing parity certificates does not alter their proofs, as established in [3].

**Proposition 3.1.** *Every rational zero of $P_N$, for $N \geq 27464$, has integer input.*

*Proof.* Over $\mathbb{Q}$, the sum of squares vanishes precisely when every quadratic row vanishes. Suppose first that $\Delta \neq 0$. For $B \geq 1$,

$$|\Delta| \leq 2B^4 + 6B^3 + 5B^2 + 1 \leq 14B^4.$$

Lemma 2.3 gives

$$B^2 \leq 5\mathcal{H}(14B^4)^{19/40} < 20\mathcal{H}B^{19/10},$$

because $14^{19} < 4^{40}$. Hence $B^{1/10} < 20\mathcal{H}$, contradicting $B \geq |a| > (40\mathcal{H})^{10}$. Thus $\Delta = 0$.

The elementary Lucas descent proved in [1, Section 4] applies to all nonzero integer $b$ and $|a| > 7$, without a parity condition. It gives

$$\log B > |a| - 3.$$

If $n - \beta \neq 0$, write $J$ for its reduced numerator. The elementary rational-height bound gives $|J| \leq 2Be^{h(\beta)}$. The order-ten divisibility and Lemma 2.3 would imply

$$B^2 \leq 5\mathcal{H}(2Be^{h(\beta)})^{19/10} < 20\mathcal{H}B^{19/10}e^{19h(\beta)/10},$$

and therefore

$$\log B < 10\log(20\mathcal{H}) + 19h(\beta) < \frac{29}{40}|a|.$$

This contradicts the Lucas bound because $|a| > 11$. Consequently $n = \beta \in \mathbb{Z}$.

Finally write $X = u/v$ in lowest terms, $v > 0$. The equality $51X^2 + 2504 \in \mathbb{Z}$ implies $v^2 \mid 51$. Since 51 is squarefree, $v = 1$, proving $X \in \mathbb{Z}$. □

## 4 Completeness and equal scales

**Proposition 4.1.** *Under the stated arithmetic premises, every integer input extends to a rational zero of $P_N$ for every $N \geq 27464$.*

*Proof.* For an integer input, $n = \beta = 51X^2 + 2504$ is an integer. Choose the remaining integer labels and elliptic witnesses by the scalar and Lucas completeness construction in [3]. The root charts admit every nonzero integer label by Lemma 2.1.

The small-prime scalar checks can be stated explicitly. For the integer $r$ above,

$$v_2(r) = \begin{cases} 3, & 4 \mid X, \\ 2, & 4 \nmid X, \end{cases} \qquad v_3(r) = 1, \qquad r \equiv 15 \pmod{17}.$$

Since $9792 = 2^6 \cdot 3^2 \cdot 17$, the local correction $(R + 9792)/R$ is a square at $2, 3, 17$. These are exactly the local conditions used in the retained scalar pole realization. The fixed avoidance at $\ell$ was already checked above.

After fixing the integer root labels, apply the retained prime-pattern construction to the distinct scales in (4). They are positive rational squares, and at least two remain since $S_0 > 12$ implies $S_0^2 \neq 1$. Equal scales have identical mathematical detectors and reuse the same witnesses. No equation requires the three scales to be distinct; in particular $S_1 = 0$ creates no exception.

The conjugate reduction-order congruences used for the comparison points remain valid for arbitrary fixed integer root labels. At the fixed small primes, choose compact local neighborhoods avoiding the finitely many poles of the rational functions, as in the source completeness proof. Their valuations are then bounded below by constants depending on the already fixed labels. The pole preparation can increase the chosen local depths after these constants have been fixed. This is the same local argument used for even input labels in [3]; it needs no uniform bound as the input varies. The real approximation and quantitative separation steps then give the simultaneous height-pole witnesses for these three scales.

Fill in the remaining arithmetic-circuit coordinates by their stated rational formulas. The retained denominator conditions hold for these completeness witnesses. The two new scale equations hold by their definitions in the next section. Hence all 342 quadratic rows vanish, as required. □

## 5 Literal specification and independent verification

The exact source system is the 389-variable table of [2], whose uncompressed SHA-256 digest is

`f7fb5bdf6a3d47275f1d78ed5e275954e2cc1f21fec8204d6a395bfc2f8b5394`.

Source identifiers are stable identifiers in that table; they are not the consecutive coordinate numbers of the new system. The complete list of all 370 coordinates is included in both new tables.

For the source root coordinates, the actual abscissas are

$$x(Q_1) = x_{82} + 2x_{83}\rho, \quad x(Q_2) = x_{84} + 2x_{85}\rho,$$
$$x(Q_3) = 2x_{86} + 2x_{87}\rho, \quad x(Q_4) = x_{88} + 2x_{89}\rho.$$

Thus

$$S_0 = x_{82} + x_{84} + 2x_{86} + x_{88}, \qquad S_1 = 2(x_{83} + x_{85} + x_{87} + x_{89}).$$

The two scale coordinates satisfy

$$S_0^2 = 4x_{343}, \qquad (S_0 + S_1)^2 = 4x_{344}.$$

For the third detector, the constructor uses a separate coordinate copy of the entire second detector, with $x_{343}$ replaced by $x_{344}$. Both nonconstant detectors therefore use the same literal normalization: their scale is four times the stored scale coordinate and their contact input is four times their stored input coordinate. The semantic scales are exactly (4).

More explicitly, for either nonconstant scale $a$, put $t = a\lambda$ and $\delta = t - 1$, and let

$$\Theta = \{0, -1/2, -1, 1/2399, -32/407\}, \qquad F(t) = \prod_{\theta \in \Theta} (t - \theta).$$

The independently expanded equations are

$$F(t)e = (a\delta)^5, \qquad \delta^3 q_1 = e, \qquad e\delta b = T^2.$$

The retained toric gates compute $q_{10} = e^{19}/\delta^3$ and $q_{40} = e^{79}/\delta^3$ on their nonzero domain. Thus every power, field product and pole input has the source normalization. At consumed contact primes the local nonvanishing argument justifies that domain.

The constructor removes the six product coordinates with source identifiers $337, \ldots, 342$ and their six equations. It also removes the twelve odd-index coordinates and their twelve sole-use equations, and the obsolete scale coordinate $x_{345}$ and its defining equation. The old input is replaced by $(X - 1)/2$, its square coordinate by $(Y - 2X + 1)/4$, and the retained square equation becomes $Y = X^2$. This converts the old label to $51X^2 + 2504$. The scalar shift is 9792. After each row is made primitive with a fixed sign, their sum of squares is expanded by exact integer arithmetic.

The independent checker performs the substitution in the reverse direction for every unchanged row. It separately checks the two new scale rows, the root-coordinate normalizations, the sole uses of deleted coordinates, all finite prime congruences and the exact coefficient expansion. All 50 equations of the third detector recover the corresponding second-detector equations under the inverse coordinate permutation; every use of its 52 coordinates is checked. The full rational normalizations displayed above are also verified as exact polynomial identities. It computes the sum of squares by triangular convolution, independently of the producer's ordered-pair expansion. A coefficient equal to one proves content one for every integer parameter; a nonzero degree-four term proves the degree is exactly four. A coefficient root bound proves the stated support for $N \geq 27464$, and direct integer evaluation gives the stated binary size for the explicit member.

| Quantity | Exact value |
|---|---|
| Total rational coordinates | 370 |
| Quadratic equations | 342 |
| Polynomial degree | 4 |
| Terms before grouping powers of $A_N$ | 5852 |
| Coordinate monomials, $N \geq 27464$ | 5837 |
| Content | 1 |
| Largest coefficient bits, $N = 27464$ | 1791357 |

Reproduction requires only Python's standard library:

```
python3 build_quartic370.py
python3 check_quartic370.py
```

The compressed exact source table is included. These finite checks establish the algebraic specification and numerical certificates. Propositions 3.1 and 4.1, together with the stated arithmetic premises, establish the rational input projection.

## References


[1] *A forced prime condition and an explicit quartic in 392 variables*, 8 November 2026, source arithmetic contract and exact circuit, https://github.com/JoseBrox/diophantine-z-q_improvements, papers/quartic/quartic392_20261108_v8.tex.

[2] *A quartic in 389 variables from local nonvanishing*, 8 November 2026, same repository, papers/quartic/quartic389_20261108_v9.tex.

[3] *A family of quartics in 317 variables from aggregate trace heights*, 8 November 2026, same repository, papers/quartic_existence/quartic317_family_20261108_v9.tex.

[4] *Effective heights of sums of elliptic abscissas*, 8 November 2026, companion paper, effective_elliptic_sums_20261108_v10.tex.